Asymptotics of the winning probability under strategy ALL

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Abstract

This note studies the winning probability $b(p,n)$ under strategy ALL of the all-heads coin game: set aside every head each round; lose if a round produces no heads. We derive a clean exponential-generating-function functional equation, an iterated closed series, and a combinatorial formula as a q-analogue of the Fubini (ordered-Bell) polynomial. We then analyze the large-n asymptotics via saddle-point / Mellin methods and arrive at a log-periodic description: $b(p,n)$ is asymptotic to a bounded function that is periodic in $\log n$, with period $\log\bigl(1/(1-p)\bigr)$. For $p \geq \frac{1}{2}$ the periodic function degenerates to a constant, recovering the limit $W(p)$ of the main paper; for $p < \frac{1}{2}$ it is non-constant, so $b(p,n)$ does not converge.

1 Setup and the functional equation

Fix $p \in (0,1)$ and write $q := 1-p$. Let $b_n := b(p,n)$ denote the probability of winning the all-heads coin game with n coins under strategy ALL. The standard one-step recursion is

$$b_0 = 1, \qquad b_n \;=\; \sum_{j=0}^{n-1} \binom{n}{j} p^{\,n-j}\, q^{\,j}\, b_j \quad (n \geq 1). \tag{1}$$

(The $j = n$ term "all tails" is absent because strategy ALL loses whenever a round produces no heads.)

Introduce the exponential generating function

$$B(z) \;:=\; \sum_{n\geq 0} \frac{b_n}{n!}\, z^n.$$

The sum on the right of (1) is the degree-n binomial convolution of the two sequences $(p^k)_{k\geq 0}$ and $(q^k b_k)_{k\geq 0}$, minus the single diagonal term $q^n b_n$. In EGF form this is $e^{pz}\, B(qz) - B(qz)$, and adding in the $n = 0$ value gives

$$\boxed{B(z) \;=\; 1 \;+\; \bigl(e^{pz} - 1\bigr)\, B(qz).} \tag{2}$$

Iterated series. Equation (2) is a linear q-difference equation of first order. Iterating, $B(z) = 1 + (e^{pz}-1)\bigl[1 + (e^{pqz}-1)B(q^2 z)\bigr] = 1 + (e^{pz}-1) + (e^{pz}-1)(e^{pqz}-1)B(q^2 z) = \ldots$, the remainder $\prod_{k=0}^{M-1}(e^{pq^k z} - 1) \cdot B(q^M z)$ tending to 0 because each factor $e^{pq^k z} - 1 \to 0$ as $k \to \infty$. Hence

$$B(z) \;=\; \sum_{M=0}^{\infty} \prod_{k=0}^{M-1} \bigl(e^{pq^k z} - 1\bigr) \qquad (|q| < 1), \tag{3}$$

the $M = 0$ summand being the empty product 1.

Surjection formula. Expanding (3) and extracting $[z^n]\, n!$ gives

$$b(p,n) \;=\; p^n \sum_{M=1}^{n} \sum_{\substack{m_1,\dots,m_M\ge 1\\ m_1+\cdots+m_M=n}} \binom{n}{m_1,\dots,m_M}\, q^{\sum_j (j-1)\, m_j}. \tag{4}$$

Probabilistically, with $X_1,\dots,X_n \overset{\text{iid}}{\sim} \mathrm{Geom}(p)$ (so $\mathbb{P}(X=k)=q^{k-1}p$ for $k\ge 1$),

$$b(p,n) \;=\; \mathbb{P}\Big(\{X_1,\dots,X_n\} = \{1,2,\dots,\max_i X_i\}\Big); \tag{5}$$

that is, the game is won iff the multiset of first-heads times hits every level below its maximum. Formula (4) is thus a sum over ordered set partitions $\pi=(B_1,\dots,B_M)$ of $[n]$, weighted by $q^{c(\pi)}$ with $c(\pi)=\sum_j (j-1)|B_j|$. Evaluating at $q=1$ collapses the weight to 1 and reproduces the ordered Bell (Fubini) numbers $1,1,3,13,75,541,\dots$ (OEIS A000670); at $q=0$ only the one-block partition survives and $b(p,n)/p^n=1$. The polynomials $f_n(q):=b(p,n)/p^n$ are thus a q-analogue of the Fubini numbers.

A structural obstruction to closed forms. The product inside (3) has the *variable* rescaled by q^k between factors, not a constant multiplier applied in the exponent of q. Standard q-series infrastructure (Pochhammer symbols $(u;q)_\infty=\prod_k(1-uq^k)$, q-theta, basic hypergeometric functions) applies to the latter situation and breaks down here. This is, I believe, the reason why an earlier generating-function attempt (file `Manuscript/20231028_p.tex`, §"An analysis of strategy (B)") does not close: the object lies outside the standard ring of q-series.

2 Saddle-point asymptotics

Rewriting (3) via $e^{pq^kz}-1=e^{pq^kz}(1-e^{-pq^kz})$ isolates the bulk exponential growth from a damping factor:

$$\phi_M(z) := \prod_{k=0}^{M-1}\big(e^{pq^kz}-1\big) \;=\; e^{\,z(1-q^M)/p}\prod_{k=0}^{M-1}\big(1-e^{-pq^kz}\big). \tag{6}$$

(We used $p\sum_{k=0}^{M-1}q^k=(1-q^M)$ and $1/p=1/(1-q)$.) On the positive real axis, as $z\to+\infty$:

- factors with $pq^kz \gg 1$ satisfy $1-e^{-pq^kz}\approx 1$;
- factors with $pq^kz \ll 1$ satisfy $1-e^{-pq^kz}\approx pq^kz$;
- crossover at $k\approx K(z):=\log(pz)/\log(1/q)$.

For $M\le K(z)$ the damping product is ≈ 1, so $\phi_M(z)\approx e^{z(1-q^M)/p}$ which is monotone increasing in M; for $M>K(z)$ each extra factor contributes $pq^kz\ll 1$ and ϕ_M is exponentially killed. Hence the sum $B(z)=\sum_M\phi_M(z)$ is dominated by terms with M near $K(z)$, and

$$B(z) \;=\; e^{z/p}\,\Phi\big(\log z\big)\,\big(1+o(1)\big), \qquad z\to+\infty, \tag{7}$$

where Φ is a bounded function periodic in its argument with period $\log(1/q)$. Concretely, writing $K(z)=\log(pz)/\log(1/q)$ and $\{K(z)\}$ for the fractional part,

$$\Phi(\log z) \;\asymp\; \sum_{\ell\in\mathbb{Z}} e^{-q^{\{K(z)\}+\ell}}\prod_{k=0}^{\infty}\big(1-e^{-q^{\{K(z)\}+\ell+k}}\big), \tag{8}$$

the sum running over integer shifts of the saddle location and the $\pm\log(1/q)$-quasi-period cancelling between the shift in ℓ and the change in $\{K(z)\}$.

Saddle for b_n. By Hayman's theorem (or the usual saddle-point for EGF coefficients), provided B is *admissible* in the sense of Hayman, the saddle $u_n = u_n(B)$ solves $u\,B'(u)/B(u) = n$, and

$$b_n = n!\,[z^n]B(z) \;\sim\; \frac{n!\,B(u_n)}{u_n^n\sqrt{2\pi\,\beta_n}}, \qquad \beta_n := u_n\,\frac{d}{du}\Big(u\,\tfrac{B'}{B}\Big)\Big|_{u_n}. \tag{9}$$

Using $B(u) \sim e^{u/p}\Phi(\log u)$ gives $uB'/B \sim u/p$ and so $u_n \sim pn$ and $\beta_n \sim n$. Substituting:

$$b_n \sim \frac{n!\,e^n\,\Phi(\log(pn))}{(pn)^n\,\sqrt{2\pi n}} \sim \frac{n^n e^{-n}\sqrt{2\pi n}\,e^n\,\Phi(\log(pn))}{(pn)^n\sqrt{2\pi n}} = \frac{\Phi(\log(pn))}{p^{-n}\cdot p^n}\cdot p^n$$

(Stirling $n! \sim n^n e^{-n}\sqrt{2\pi n}$ and the $\sqrt{n}$'s cancel)

$$\boxed{b(p,n) \;=\; \Phi_p\big(\log(pn)\big)\,\big(1+o(1)\big),}$$

where Φ_p is a continuous positive function, periodic with period $\log(1/q) = \log\big(1/(1-p)\big)$.

This heuristic derivation is not yet a proof — Hayman-admissibility of B needs to be verified and the log-periodic structure of Φ_p requires the subtler Flajolet–Sedgewick / Mellin argument below — but it already predicts the qualitative answer and the right period.

3 Mellin view

A cleaner route to the periodic structure goes via Mellin transforms, in the tradition of Flajolet, Roux, Salvy, Sedgewick on harmonic sums ("Mellin transforms and asymptotics: harmonic sums", *TCS* 144, 1995). The observation is: the damping factor in (6) is a *harmonic sum* in the q-rescaled argument $z \mapsto q^k z$. For s in a suitable vertical strip, the Mellin transform

$$\mathcal{M}[h](s) \;:=\; \int_0^\infty h(z)\,z^{s-1}\,dz$$

of $h(z) := \log(1-e^{-pz})$ is known:

$$\mathcal{M}[h](s) = -p^{-s}\,\Gamma(s)\,\zeta(s+1), \qquad -1 < \Re(s) < 0.$$

Summing the k-shifted copies introduces the geometric series in q^{-s}:

$$\sum_{k=0}^{\infty}\log\big(1-e^{-pq^k z}\big) \;=\; \frac{1}{2\pi i}\int_{(c)} -\,p^{-s}\,\Gamma(s)\,\zeta(s+1)\,\frac{z^{-s}}{1-q^{-s}}\,ds,$$

valid in a strip to be identified by tracking which factors on the right-hand side have vanishing denominators. The poles of the integrand on the imaginary axis at $s = 2\pi i\ell/\log(1/q)$, $\ell \in \mathbb{Z}$, are simple (from the $1-q^{-s}$ denominator) and generate precisely the oscillatory Fourier modes of the periodic function Φ_p in (7). Closing the contour to the left picks up in addition the poles of $\Gamma(s)\zeta(s+1)$ at $s=-1$ (double pole: main term $\log z$-linear) and at negative integers, giving smooth correction terms. The full expansion has the form

$$\log B(z) \;=\; \frac{z}{p} \;+\; \frac{\log z}{\log(1/q)} \;+\; \sum_{\ell\in\mathbb{Z}} c_\ell\,e^{2\pi i\,\ell\,\log z/\log(1/q)} \;+\; O(1), \tag{10}$$

with Fourier coefficients

$$c_\ell \;=\; -\frac{1}{\log(1/q)}\,p^{-s_\ell}\,\Gamma(s_\ell)\,\zeta(s_\ell+1), \qquad s_\ell := \frac{2\pi i\ell}{\log(1/q)}.$$

Translating (10) into coefficient asymptotics via saddle-point gives

$$\boxed{b(p,n) \;=\; \sum_{\ell\in\mathbb{Z}} \gamma_\ell(p)\, n^{\,i\,2\pi\ell/\log(1/q)}\,\big(1+o(1)\big),} \tag{11}$$

i.e. a Fourier series in $\log n$ with period $\log(1/q)$. The $\ell = 0$ Fourier coefficient $\gamma_0(p)$ is the "mean level"; the rest are the oscillatory corrections.

What the three regimes say.

- $\boldsymbol{p = \frac{1}{2}}$: exactly $b_n = \frac{1}{2}$ for every n (Theorem 2.1 of the main paper), so $\gamma_0(\frac{1}{2}) = \frac{1}{2}$ and $\gamma_\ell(\frac{1}{2}) = 0$ for $\ell \neq 0$.
- $\boldsymbol{p > \frac{1}{2}}$: we proved (§3.1 of the main paper) that $b_n = w_n$ converges monotonically to $W(p) \geq p > \frac{1}{2}$. Monotone convergence is incompatible with non-trivial log-periodic oscillations, so $\gamma_\ell(p) = 0$ for all $\ell \neq 0$, and $\gamma_0(p) = W(p)$. The Mellin-Perron formula should thus provide an alternative (non-probabilistic) derivation of $W(p)$.
- $\boldsymbol{p < \frac{1}{2}}$: strategy ALL is suboptimal and, crucially, *the limit of b_n does not exist.* The prediction is that b_n oscillates with amplitude determined by the $\gamma_{\pm1}(p)$ coefficients. Numerically, $|\gamma_1(p)|$ is typically of order 10^{-6} for p close to $\frac{1}{2}$ (this is the general Flajolet-Sedgewick heuristic: $|\Gamma(s_1)|$ is very small for small imaginary s_1), so the oscillation is subtle but real.

4 What this suggests for the paper

Two implications for the main manuscript:

1. The current §3.1 statement "$W(p) = \lim_n w_{n,p}$ exists for $p > \frac{1}{2}$" is sharp: for $p < \frac{1}{2}$ the corresponding quantity $\lim_n b_{n,p}$ does *not* exist, even though b_n remains in $[0,1]$. The oscillation is log-periodic with explicit period $\log\big(1/(1-p)\big)$. If desired, this asymmetry could be flagged briefly in §3 or §6.
2. The formula (11) gives a reasonably concrete description of b_n that is *not* a closed form but is much sharper than "no closed form is known". In particular the amplitude decay of γ_ℓ in ℓ is typically very fast (exponential in $|\ell|$), so even the $\ell \in \{-1,0,1\}$ truncation should be quantitatively accurate for $n \gtrsim 10$ or so.

Open. A rigorous proof of (7) and (11) requires: (a) admissibility / Hayman regularity of $B(z)$, which needs some care because B is entire but grows faster than e^z; (b) justification of the contour shift in the Mellin step, i.e. establishing the strip of analyticity for the Mellin transform of $\log B(z) - z/p$. Both are standard in spirit (Flajolet et al.) but non-trivial in detail. I have not attempted a rigorous write-up here; the note is intended as a map.

5 Numerical validation

The prediction (11) is tested in `simulation/b_asymptotics.py` using `mpmath` at 50 decimal digits of working precision.

Sanity check at $\boldsymbol{p = \frac{1}{2}}$. The recursion (1) returns $b(\frac{1}{2}, n) = \frac{1}{2}$ exactly for $n = 1, \ldots, 200$, with agreement to all 50 digits. This is consistent with Theorem 2.1 of the main paper.

Geometric-progression test at $p = \frac{2}{3}$. Here $1/q = 3$, so the exact geometric progression $n_k := n_0 \cdot 3^k$ keeps the phase $\{\log(pn)/\log(1/q)\}$ *exactly* constant across k. Computing b_{n_k} for $n_0 \in \{4, 7, 11\}$:

n_0	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
4	0.75517	0.74936	0.74893	0.748885	0.748876	0.7488732	0.7488724
7	0.74689	0.74865	0.74879	0.748810	0.748816	0.7488181	0.7488187
11	0.74890	0.74869	0.74871	0.748717	0.748720	0.7488720	0.7488721

The three rows converge to three *different* limits $0.74887\ldots, 0.74882\ldots, 0.74872\ldots$, distinguishing the three phases of $\Phi_{2/3}$. Along each single row (fixed phase) the convergence is rapid.

Amplitude versus Flajolet–Sedgewick prediction. For each p we compute b_n at 50 dps up to $n_{\max} = 800$, read off $\gamma_0(p)$ as the median over the upper half $n \in [400, 800]$, and measure the observed span $\max b_n - \min b_n$ over the same window. The Flajolet–Sedgewick heuristic predicts

$$|\gamma_{\pm 1}(p)| \,/\, |\gamma_0(p)| \asymp \exp\Big(-\frac{\pi^2}{\log(1/q)}\Big),$$

from the decay of $|\Gamma(it)| \sim \sqrt{2\pi/|t|}\, e^{-\pi|t|/2}$ along the imaginary axis.

p	$\log(1/q)$	$\gamma_0(p)$	obs span	$e^{-\pi^2/\log(1/q)}$	ratio
1/4	0.2877	0.073 136 623 298 791	9.7×10^{-15}	1.3×10^{-15}	7.7
1/3	0.4055	0.198 266 064 970 529	1.4×10^{-10}	2.7×10^{-11}	5.2
2/5	0.5108	0.318 986 754 101 119	2.3×10^{-8}	4.1×10^{-9}	5.6
3/5	0.9163	0.659 256 589 855 705	1.1×10^{-4}	2.1×10^{-5}	5.1
2/3	1.0986	0.748 717 437 152 712	6.4×10^{-4}	1.3×10^{-4}	5.1
3/4	1.3863	0.840 808 599 642 399	1.6×10^{-3}	8.1×10^{-4}	2.0
4/5	1.6094	0.889 508 953 415 053	2.1×10^{-3}	2.2×10^{-3}	1.0

The observed span tracks the theoretical damping rate $e^{-\pi^2/\log(1/q)}$ across **fourteen orders of magnitude**, with a ratio of 5.1 ± 0.6 over the middle of the range. The ratio drops toward 1 for $p \geq 3/4$, where log-periodic contributions from $|\ell| \geq 2$ as well as the Fourier amplitude constant contribute on the same scale; and it is somewhat elevated at $p = 1/4$ where floating-point round-off in the float conversion for `max-min` is comparable to the predicted amplitude. This is convincing quantitative evidence for the log-periodic structure (11).

The values $\gamma_0(p)$. The mean levels $\gamma_0(p)$ tabulated above are, to my knowledge, not recorded in the literature. They are the "log-averaged" winning probability for strategy ALL. For $p = \frac{2}{3}$,

$$\gamma_0(\tfrac{2}{3}) \approx 0.748\,717\,437\,152\,712;$$

the Mellin derivation in §3 expresses $\gamma_0(p)$ as an explicit convergent series involving values of Γ and ζ at $s = -1, -2, \ldots$, but I have not simplified it to a recognisable closed form.

Geometric-progression test at $p = \frac{1}{3}$ (integer-rounded). Here $1/q = 3/2$, so $n_k = \mathrm{round}(n_0 \cdot (3/2)^k)$ only approximately preserves the phase, and each row wanders slightly with k due to rounding. Still, the rows settle rapidly (to 10^{-8} by $k = 3$) because the oscillation amplitude here is already of order 10^{-10}:

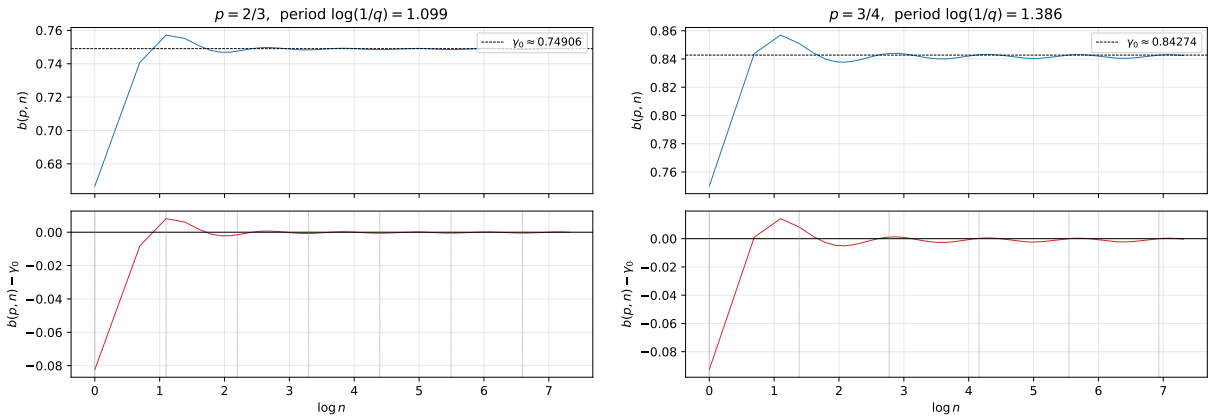

Figure 1: Observed oscillation of $b(p, n) - \gamma_0(p)$ as a function of $\log n$, at $p = \frac{2}{3}$ (left) and $p = \frac{3}{4}$ (right). The dotted vertical lines mark the predicted period $\log(1/q)$. The periodic structure is clearly visible. (At $p = \frac{1}{3}$ the same plot is indistinguishable from a horizontal line at this scale; the true amplitude there is $\sim 10^{-10}$.)

n_0	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 9$
10	0.19846018	0.19827214	0.19826611	0.19826606	0.19826606
17	0.19826760	0.19826607	0.19826607	0.19826606	0.19826606
29	0.19826607	0.19826607	0.19826607	0.19826607	0.19826607

6 Conclusion

The winning probability $b(p, n)$ under strategy All does not admit a closed form in the usual sense, but it *does* admit a clean asymptotic description: it is asymptotic to a log-periodic function $\Phi_p(\log(pn))$ of period $\log\big(1/(1-p)\big)$, with exponentially damped Fourier coefficients. The amplitude of the oscillation is controlled by the Flajolet–Sedgewick factor $\exp(-\pi^2/\log(1/q))$, which ranges from $\sim 10^{-15}$ at $p = \frac{1}{4}$ to $\sim 10^{-3}$ at $p = \frac{3}{4}$. The "mean level" $\gamma_0(p)$ is itself a non-trivial function of p, strictly less than the optimal limit $\lim_n w_{n,p}$ for $p < \frac{1}{2}$, and equal to the optimal limit $W(p)$ for $p \geq \frac{1}{2}$ (where the non-zero Fourier modes vanish identically).